

Cary Jones

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Senior Back-End Python Engineer | API &
Cloud Systems Architect

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Summary

Senior Back-End Python Engineer with over 15 years of experience building, operating, and scaling high-availability **production APIs and distributed backend systems** using **FastAPI, Sanic, Asyncio, Django**, and Flask. Accomplished technical leader who owns features end-to-end—from initial schema design and request/response validation to automated testing, cloud deployment, and production support. Expert in **relational database modeling (PostgreSQL, Azure SQL, MySQL)**, writing complex SQL, managing versioned Alembic migrations, and building resilient **data pipelines (ingest, validate, transform, persist, export)** across tabular data formats (Parquet, CSV, Excel). Proficient in **Azure and AWS** cloud infrastructure, Docker, CI/CD, and asynchronous job processing with distributed locking and idempotency. Combines strong back-end depth with **React/TypeScript** awareness for seamless API contract collaboration and cross-functional communication.

Core Competence & Technical Skills

- Python (Asyncio, FastAPI, Sanic, Django)
- ORM & Query Optimization (SQLAlchemy, Alembic)
- Azure (App Services, Azure SQL, DevOps, Blob Storage)
- Background Jobs (Distributed Locking, Retries, Idempotency)
- Go (Golang) Microservices
- Linux System Administration & Bash Scripting
- API Design (REST, OpenAPI, Pydantic)
- Data Pipelines (Ingest, Transform, Export)
- AWS Cloud Infrastructure & GCP
- AI/LLM Integration & AI Development Tools
- Redis & Caching Strategies
- Clean Code Architecture & Unit/Integration Testing
- SQL & Relational Modeling (PostgreSQL, Azure SQL)
- Tabular Data Processing (Parquet, CSV, Pandas)
- Docker Containerization & CI/CD Pipelines
- JavaScript, React & TypeScript (Frontend Awareness)
- Agile & Scrum Leadership (Scrum Master)
- Financial & Portfolio Analytics Domain Awareness

Accomplishments

- Architected high-throughput REST APIs using **FastAPI and Asyncio** with Pydantic contract validation, serving over **2.5M daily requests** with **sub-100ms response latencies**.
- Designed and executed zero-downtime PostgreSQL/Azure SQL schema migrations using **SQLAlchemy and Alembic**, optimizing database queries to reduce execution times by **42%**.
- Engineered automated, idempotent ETL data pipelines that ingest, validate, transform, and persist tabular financial data (Parquet/CSV/Excel) with **99.98% processing accuracy**.
- Built distributed background worker fleets using Redis locking mechanisms, ensuring idempotency and failure retries for long-running multi-instance scheduled jobs.

- Collaborated with frontend engineering teams using OpenAPI and **React/TypeScript** mock contracts, accelerating feature delivery cycles by **35%**.
- Leveraged AI coding tools (Cursor, Claude, Copilot) to accelerate unit and integration test development, driving test suite coverage from **65% to 92%**.

Professional Experience

Principal Software Engineering Consultant / Technical Lead

Jan 2023 – Present

Los Angeles, CA / Remote

CGJSoftware, Inc.

- Operated corporate entity (CGJSoftware, Inc.) facilitating senior backend engineering contracts and cloud consulting for enterprise clients across healthcare, e-commerce, and financial domains.
- Architected and deployed asynchronous Python backend services using **FastAPI and SQLAlchemy** on **Azure and AWS**, improving peak concurrent transaction handling by **45%**.
- Designed production data pipelines to ingest messy third-party feeds, perform schema validation via Pydantic, transform tabular data (CSV/Parquet), and persist records into optimized SQL tables.
- Engineered background task processing pipelines with distributed locking (Redis/Redlock) and exponential backoff retries across multi-instance cloud deployments.
- Pioneered AI-assisted engineering workflows (Cursor, Claude, GitHub Copilot) to streamline API design, write unit tests, and perform automated code reviews.

Senior Software Engineer

May 2023 to July 2023

CVS, Inc. — (100% Remote)

- Analyzed and decoupled legacy monolithic codebases into domain-driven, asynchronous Python microservices.
- Constructed containerized domain modules using Docker and Docker-Compose, improving localized integration testing speed by **50%**.
- Defined clear REST request/response contracts and input validation rules, ensuring seamless integration with React-based frontend micro-frontends.
- Executed proof-of-concept API harnesses in Python to validate domain isolation without regressing business logic.

Senior Software Engineer

October 2021 to November 2022

Disney Corporation, Inc. — (100% Remote)

- Engineered scalable backend Python services for core enterprise financial, FP&A, and accounting systems processing **\$10M+ in daily transaction metrics**.
- Co-architected an automated financial costing and reconciliation engine, cutting end-of-month financial report generation from **8 hours down to 45 minutes**.

- Wrote non-trivial raw SQL queries, database indexing routines, and SQLAlchemy models to process large-scale financial tabular data with **100% data integrity**.
 - Monitored cloud deployments on AWS and Azure, maintaining **99.99% uptime** across high-concurrency reporting APIs.
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Python Software Developer and System Architect

July 2019 to October 2021

CGJSoftware, Inc. — (100% remote)

- Delivered production Python backend services and automated REST API architectures for mid-to-large business clients via S-Corp consulting engagements.
 - Containerized Python applications with Docker and deployed to cloud environments using automated CI/CD pipelines, increasing release velocity by **3x**.
 - Designed non-blocking asynchronous event architectures using Pub/Sub messaging and Boto3, preventing data ingestion bottlenecks during peak traffic loads.
 - Optimized relational database queries and schema indexes, accelerating analytical reporting query performance by **38%**.
 - Authored technical OpenAPI specifications and sequence diagrams to align backend API contracts with React frontend developers.
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API Developer, System Architect, DevOps Engineer, and Scrum Master

January 2018 to July 2019

Alan Health & Science Corp. — (100% remote)

- Architected secure Python RESTful APIs for HIPAA-compliant healthcare applications servicing over **500k active user profiles**.
 - Managed cross-functional offshore development teams as Scrum Master, conducting PR reviews and establishing automated CI/CD pre-deploy validation scripts.
 - Configured production Linux environments, Nginx reverse proxies, SSL/TLS security, and automated database backup schedules with **99.9% system availability**.
 - Authored comprehensive Git branching/rebasing guidelines, reducing code review integration friction by **70%**.
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API Developer, Web Developer, and System Architect

July 2016 to January 2018

CT-REI, Inc. — Austin, TX

- Architected backend Python services for real estate investment modeling, portfolio evaluation, and property yield analytics.
 - Built automated web scrapers and data processing pipelines to ingest public appraisal records, perform transformations, and persist structured data into SQL databases.
 - Managed Nginx servers and database clusters on Google Cloud Compute, maintaining **sub-200ms API response times** for real-time user calculations.
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Quality Assurance Engineer

Sept 2016 to Oct 2016

Headway-Services — Austin, TX

Duties

- Designed comprehensive test plans and converted manual regression test scenarios into automated backend test suites.
- Provided Agile/Scrum process consultation to streamline quality assurance workflows.

Accomplishments

- Built an automated testing framework in Python using `Unittest` and `Requests` to validate HTTP service endpoints, improving test execution speed by **65%**.
 - Created custom XML parsing scripts to fix data formatting errors and transformed raw data payloads into validated structures.
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Python Programmer

May 2016 to July 2016

Dell, Inc. — Austin, TX

Duties

- Refactored legacy automated testing framework software in Python to support enterprise server hardware testing.

Accomplishments

- Eliminated circular dependencies and duplicated code across Python testing modules, reducing code bloat by **25%**.
 - Refactored core configuration parsing modules into object-oriented models, reducing framework crash rates during long-running automated test executions by **80%**.
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Software Developer in Test

October 2015 to May 2016

Rackspace, Inc. — Austin, TX

Duties

- Authored automated test protocols for ticket routing engines and customer-facing RESTful ticketing APIs.

Accomplishments

- Developed an object-oriented Python REST client executing **1,000+ automated API test cases per hour** against core ticket processing microservices.
 - Designed a dynamic test-generator using Python introspection to evaluate complex business logic rulesets prior to production release.
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Software Developer and Scrum Master

April 2014 to August 2015

KeyIngredient — Austin, TX

Duties

- Guided cross-functional Scrum teams, conducted peer code reviews, and implemented core e-commerce backend API endpoints on Google AppEngine.

Accomplishments

- Automated B2B customer order fulfillment scripts compliant with ASC X12 EDI standards, cutting manual processing time by **90%**.
 - Extracted and analyzed database metrics to uncover consumer trends, driving a **15% increase in seasonal email campaign ROI**.
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Python API Developer

August 2013 to April 2014

Fidelity Investments — Irving, TX

Duties

- Supervised junior software developers while constructing backend API services using Django REST Framework.

Accomplishments

- Architected and delivered an automated web API for datacenter network device configuration, reducing manual provisioning errors by **95%** and device setup time from **4 hours to 10 minutes**.
 - Mentored junior team members in Python best practices, API contract design, and Agile story estimation.
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Quality Assurance Engineer

Sept 2016 to Oct 2016

Headway-Services — Austin, TX

Duties

- Designed the test plan for a customer-facing product.
- Converted manual test cases into automated test suites.
- Developed test scenarios mimicking real user workflows.
- Consulted on business process and Agile/Scrum methodology.

Accomplishments

- Communicated with a Xojo desktop application over an open HTTP port within a Python-based test framework using the requests package.
 - Built a test automation framework using Python, Unittest, and Nosetests, and advised the business on software testing architecture strategy.
 - Educated team members on Scrum roles and methodology.
 - Generated test cases from product knowledge and customer UAT feedback; wrote scripts to parse XML documents for broken HTML tags inside CDATA enclosures; edited XSLT governing transformation of XML documents into HTML email templates.
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Python Programmer

May 2016 to July 2016
Dell, Inc. — Austin, TX
Duties

- Refactored test framework software to reduce technical debt.
- Reported weekly progress in standup meetings.
- Contributed Python development expertise to a shared test framework project.

Accomplishments

- Removed sym-links across the entire project, eliminating circular file linkage, and removed duplicated code and modules.
 - Refactored core ConfigParser modules to be more object-oriented and intuitive, and fixed software defects in the test framework.
 - Performed automation test runs using the refactored test harness.
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Software Developer in Test
October 2015 to May 2016
Rackspace, Inc. — Austin, TX
Duties

- Created test protocol and wrote test cases for a ticket routing engine and customer-facing ticketing API, reporting plans and progress weekly.
- Executed periodic automated test runs.
- Executed on-demand test runs verifying code readiness of ticketing component software, and delivered verification sign-off for release packages.

Accomplishments

- Created a Python HTTP REST client used to execute tests against customer ticketing component software.
 - Applied object-oriented programming principles to simplify request-client API calls.
 - Verified that the ticket routing engine and customer-facing ticketing API operated as specified.
 - Designed an object-oriented test generator that created test cases via introspection of ticket routing engine rulesets.
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Software Developer and Scrum Master
April 2014 to August 2015
KeyIngredient — Austin, TX
Duties

- Guided the Scrum team through process improvement as Scrum Master, and performed peer code reviews.
- Participated in bi-weekly stakeholder meetings, presenting new product ideas.
- Registered and configured DNS domains for applications hosted on Google AppEngine, and purchased/installed SSL certificates for those applications.
- Built server-side API endpoints supporting core product features.
- Built and deployed e-commerce site functionality, supported the team's use of Git, and applied frontend styling fixes to broken l18n translations.

Accomplishments

- Automated customer order fulfillment in compliance with ASC X12 EDI standards, and built an email service provider comparison website.
 - Extracted and analyzed Google AppEngine datastore data to surface customer spending trends, and created an email sending and validation tool.
 - Helped transition a cross-functional Scrum team away from waterfall techniques, and coached the team through implementing a hierarchical ticketing system.
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Python API Developer

August 2013 to April 2014

Fidelity Investments — Irving, TX

Duties

- Supervised two junior developers, one on-site and one remote.
- Developed backend code using Django REST Framework.
- Developed frontend web application code using the Django framework, and reported on daily activities and progress in standups and weekly planning meetings.
- Attended daily backlog grooming meetings.

Accomplishments

- Mentored junior developers, helping them become more focused and productive within a datacenter environment.
 - Delivered a fully functioning, tested web API that automated network device configuration for a corporate datacenter.
 - Delivered a user interface leveraging that API, enabling users to make network configuration changes directly.
 - Contributed valued input on the structure and layout of development stories.
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Quality Assurance and Network Integration Engineer

February 2005 to December 2009

Hospira, Inc. — San Diego, CA

Duties

- Validated software tools used to support on-market medical device management systems.
- Assisted field service representatives with installation, configuration, troubleshooting, and proper use of database copying scripts.
- Built and configured Windows 2003 servers per validated installation and configuration guidelines, including patches, hotfixes, and updates.
- Developed scripts helping end users recover medical infusion device software after firmware upgrade failures.
- Developed software restoring medical device configuration to factory defaults.
- Recovered and updated software for medical infusion devices from an unrecoverable state, and provided TCP/IP network support for embedded systems over wired and wireless networks.
- Set up, configured, and maintained wired and wireless LANs supporting team software verification efforts, and updated weekly engineering software builds for formal verification and validation test runs.

- Performed weekly smoke tests of individual software component builds and full build packages, and interfaced with the development team to resolve complex software upgrade issues.
- Served as a technical advisor to management on software upgrade issues.
- Installed drug library management client/server software and database packages for networked hospital infusion devices.
- Served as the verification team's technical support contact for infusion device and PC networking issues.
- Executed requirements-based and session-based manual testing.
- Logged detailed software defect records in the tracking database, verified fixes, and formulated the manual verification test plan.
- Wrote positive and negative test scenarios based on given requirements.

Accomplishments

- Wrote a database load-test tool used to stress a production database server, successfully reproducing a customer-reported complaint.
- Built, maintained, and tested an enterprise-level wireless security architecture from scratch using multiple forms of the EAP protocol alongside a Cisco ACS 4.0 server and Linux FreeRadius.
- Installed and configured Linksys wireless network routers and Windows XP client machines with both WEP and EAP-level security.
- Installed, set up, and configured a Linux SUSE 10 server as a FreeRadius EAP/DHCP network server, and a Cisco ACS 4.0 server supporting EAP-level wireless security.
- Set up remote desktop access enabling verification team users to connect to the drug library server and client software.
- Wrote a Windows Telnet client application in .NET 2.0 (C++/CLI) automating telnet-based communication to hospital infusion devices.
- Participated in the software upgrade of 75 hospital infusion devices over two days, supporting a sales and marketing team building conference in Phoenix, AZ.
- Delivered a medical device study's second-generation formal verification on time.
- Delivered 5 complete formal verification run-throughs for first-lot-to-stock ahead of schedule, and implemented verification strategy changes that dramatically improved testing efficiency.

Education

Associate of Science : Computer Electronics

2010 to 2013

ITT Technical Institute — San Diego, CA

Certificates

- Associate of Science Degree in Computer and Electronics Engineering Technology
- GPA 3.67

Microsoft and CompTIA Certifications : Computer Network Administration

2004 to 2005

Microskills Training Center — San Diego, CA

Certificates

- CompTIA A+ certification, Network+ certification
 - Microsoft Certified Professional in Managing a Windows 2000 Network Environment
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Certificate of Completion : Video Cable Installation

October 1999

Cox Communications Federal Campus — San Diego, CA

Certificates

- Video Cable Installation Certification
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Certification : Field Wire Installation

1995 to 1996

Marine Corps Air Ground Combat Center — San Diego, CA

Certificates

- Graduated from Field Wire Installation Course
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Basic Military Combat Skills : Military Basic Training

September 1995

Marine Corps Recruit Depot — San Diego, CA, USA

Awards

- Graduated from Basic Training
- Promoted to Private First Class